The interplay of consituent order and givenness in German and Korean: An Expectation-based account

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Goals

Test "Expected-First" Strategy: Do initial NP properties (discourse accessibility / grammatical role) drive processing?
Investigating "Implied" Status: Does it pattern with "Given," "New," or take intermediate position?
Cross-Linguistic Impact: How are expectations shaped by language specific properties such as case marking and argument drop?

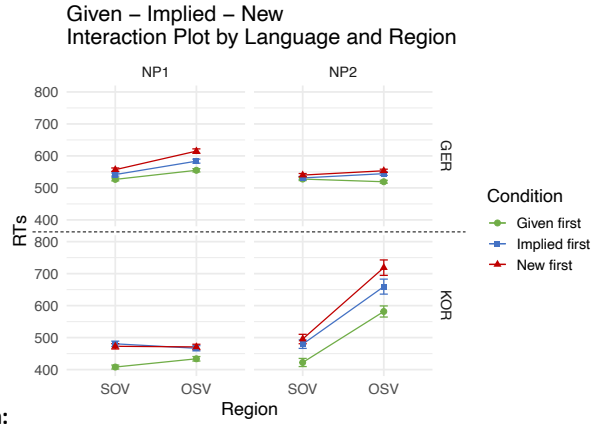
Background

Word Order (WO): Non-canonical orders increase processing effort (Bahlmann et al., 2007; Schlesewsky et al., 2003, Kim, 1992; Lee and Cho, 2003; Son, 2001)
Information Status (IS): Comprehenders prefer "Given-before-New" (Kaiser & Trueswell, 2004; Krifka & Musan, 2012; Arnold et al., 2013)
Expectation & Surprisal: Predictable structures yield lower processing costs (Hale, 2001; Levy, 2008)
Typology: German uses clear case marking; Korean allows ambiguous topic-drop

Design

		Ein Bäcker ging auf ein Konzert. Ich habe gesehen, dass ... *A baker went to a concert. I saw that ...*	한 제빵사가 콘서트에 갔다. 나는 ... *A baker went to a concert. I ...*
Givenness	WO	German Sentence Continuation	Korean Sentence Continuation
G-first	SOV	... der Bäcker dort gestern {I:den Musiker ; N:einen Piloten} ... *... the baker [SUBJ] there yesterday {I: the musician ; N: a pilot} [OBJ] ...*	... 제빵사가 어제 그곳에서 {I:음악가를 ; N:조종사를} ... *... a baker [SUBJ] there yesterday {I: a musician ; N: a pilot} [OBJ] ...*
I-first	SOV	... ein Musiker dort gestern {G:den Bäcker ; N:einen Piloten} ... *... a musician [SUBJ] there yesterday {G:the baker ; N:a pilot} [OBJ] ...*	... 음악가가 어제 그곳에서 {G:제빵사를 ; N:조종사를} ... *... a musician [SUBJ] there yesterday {G:a baker ; N:a pilot} [OBJ] ...*
N-first	SOV	... ein Pilot dort gestern {G:den Bäcker ; I:den Musiker} ... *... a pilot [SUBJ] there yesterday {G:the baker ; I:the musician} [OBJ] ...*	... 조종사가 어제 그곳에서 {G:제빵사를 ; I:음악가를} ... *... a pilot [SUBJ] there yesterday {G:a baker ; I:a musician} [OBJ] ...*
G-first	OSV	... den Bäcker dort gestern {I:der Musiker ; N:ein Pilot} ... *... the baker [OBJ] there yesterday {I:the musician ; N:a pilot} [SUBJ] ...*	... 제빵사를 어제 그곳에서 {I:음악가가 ; N:조종사가} ... *... a baker [OBJ] there yesterday {I:a musician ; N:a pilot} [SUBJ] ...*
I-first	OSV	... den Musiker dort gestern {G:der Bäcker ; N:ein Pilot} ... *... the musician [OBJ] there yesterday {G:the baker ; N:a pilot} [SUBJ] ...*	... 음악가를 어제 그곳에서 {G:제빵사가 ; N:조종사가} ... *... a musician [OBJ] there yesterday {G:a baker ; N:a pilot} [SUBJ] ...*
N-first	OSV	... einen Piloten dort gestern {G:der Bäcker ; I:der Musiker} ... *... a pilot [OBJ] there yesterday {G:the baker ; I:the musician} [SUBJ] ...*	... 조종사를 어제 그곳에서 {G:제빵사가 ; I:음악가가} ... *... a pilot [OBJ] there yesterday {G:a baker ; I:a musician} [SUBJ] ...*
		... angesprochen hat. *... talked to.*	... 부르는 것을 목격했다. *... call out to witnessed.*

Results



		NP1		Effect	NP2	
GER		**β = 44.50**	**z = 5.3**	WO	β = 5.66	z = 0.8
		β = –26.64	**z = –3.8**	I vs G	**β = –15.31**	**z = –2.7**
		β = 26.90	**z = 3.4**	I vs N	β = 8.03	z = 1.2
		β = –14.16	z = –1.3	WO : I vs G	**β = –21.47**	**z = –2.3**
		β = 14.19	z = 1.3	WO : I vs N	β = 0.49	z = 0.0
		NP1		**Effect**	**NP2**	
KOR		β = 3.18	z = 0.4	WO	**β = 187.80**	**z = 5.2**
		β = –56.84	**z = –5.5**	I vs G	**β = –65.44**	**z = –3.3**
		β = –0.79	z = –0.1	I vs N	β = 40.70	z = 1.7
		β = 41.26	**z = 2.7**	WO : I vs G	β = –23.23	z = –0.5
		β = 12.08	z = 0.8	WO : I vs N	β = 44.38	z = 1.2

German:
IS: Graded processing costs at NP1 (Given < Implied < New)
WO: Effects at NP1 (Subject < Object)
Interaction: WO and IS interaction at NP2 (Given < Implied = New for OSV)

Korean:
IS: Implied treated as New (Given < Implied = New) at NP1 & NP2
WO: Main Effect only at NP2 (SOV < OSV)
Interaction: WO and IS interaction at NP1 (Given < Implied = New mainly for SOV with consistent Main Effect of IS)

Discussion

While both languages support a **expectation-driven comprehension system** combining syntactic and information structural expectancies, their surface manifestations differ based on language-specific resources:

Commonality
In both systems, processing cost increases when NP1 violates discourse (manifest on NP1 for both GER & KOR), or grammatical expectations (manifest on NP1 for GER & NP2 for KOR)

Differences
IS: German treats "Implied" as a distinct intermediate, while Korean treats it more as "New"
Timing: German identifies roles immediately (via case); Korean shows delayed effects due to structural uncertainty (topic-drop)

References

Arnold et al., (2013), WIREs Cognitive Science; Bahlmann et al., (2007), Human Brain Mapping; Kaiser & Trueswell, (2004), Cognition; Kim, (1992), The University of Wisconsin-Madison; Krifka & Musan, (2012), In The Expression of Information Structure; Lee & Cho, (2003), Studies in Generative Grammar; Schlesewsky et al., (2003), Brain and Language; Son, (2001), The University of Wisconsin-Madison.

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